

Names :

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Why can Hill Climbing get stuck even when a valid timetable exists?

Hill Climbing can get stuck because it only moves to better neighbors and may stop at a **local minimum** even if a better timetable exists elsewhere.

How does Simulated Annealing escape local minima?

Simulated Annealing escapes local minima by sometimes **accepting worse solutions with a probability**, allowing it to explore more states.

What happens when cooling is too fast?

If **cooling is too fast**, the algorithm stops exploring quickly and may **miss the optimal solution**.

In real universities, why might accepting temporary conflicts be useful?

In real universities, accepting temporary conflicts can help **explore more scheduling possibilities before finding the best final timetable**.

Name	▲	Type	Size	Value
conflict_pairs		set	7	{{(0, 1), (2, 3), (4, 5), (1, 2), (3, 4), (0, 5), (1, 4)}}
final_score		int		0
final_state		list	6	[0, 1, 0, 1, 0, 2]
i		int		0
initial_state		list	6	[0, 1, 1, 0, 0, 1]
num_courses		int		6
num_slots		int		3
runs		int		20
success		int		20
total_conflicts		int		0

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conflict_pairs	set	7	{{(0, 1), (2, 3), (4, 5), (1, 2), (3, 4), (0, 5), (1, 4)}}
final_score	int		0
final_state	list	6	[0, 1, 0, 1, 0, 2]
i	int		0
initial_state	list	6	[0, 1, 1, 0, 0, 1]
num_courses	int		6
num_slots	int		3
runs	int		20
success	int		20
total_conflicts	int		0

```
===== RESTART: C:\Users\studentfcs.ABLOCK\Desktop\testlab2.py =====
Hill Climbing Results
Success rate: 20 / 20
Average final conflicts: 0.0

Simulated Annealing (Fast Cooling)
Success rate: 20 / 20
Average final conflicts: 0.0
Average steps: 15.45

Simulated Annealing (Slow Cooling)
Success rate: 20 / 20
Average final conflicts: 0.0
Average steps: 18.2

===== RESTART: C:\Users\studentfcs.ABLOCK\Desktop\testlab2.py =====
Hill Climbing Results
Success rate: 18 / 20
Average final conflicts: 0.1

Simulated Annealing (Fast Cooling)
Success rate: 20 / 20
Average final conflicts: 0.0
Average steps: 25.35

Simulated Annealing (Slow Cooling)
Success rate: 20 / 20
Average final conflicts: 0.0
Average steps: 15.8

===== RESTART: C:\Users\studentfcs.ABLOCK\Desktop\testlab2.py =====
Hill Climbing Results
Success rate: 20 / 20
Average final conflicts: 0.0

Simulated Annealing (Fast Cooling)
Success rate: 20 / 20
Average final conflicts: 0.0
Average steps: 18.35

Simulated Annealing (Slow Cooling)
Success rate: 20 / 20
Average final conflicts: 0.0
Average steps: 20.15

===== RESTART: C:\Users\studentfcs.ABLOCK\Desktop\testlab2.py =====
Hill Climbing Results
Success rate: 20 / 20
Average final conflicts: 0.0

Simulated Annealing (Fast Cooling)
Success rate: 20 / 20
Average final conflicts: 0.0
Average steps: 11.8

Simulated Annealing (Slow Cooling)
Success rate: 20 / 20
Average final conflicts: 0.0
Average steps: 16.7
```